WORKED SOLUTION

STEP III 1987, Question 5

Pure • Modular Arithmetic Ciphers

STEP III 1987, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q5

The key idea

The code is two separate shift ciphers, one for the even positions (key n_0) and one for the odd positions (key n_1), each with an extra shift of j. Strip off the known shift j first. What remains is each half of the message shifted by one unknown constant, and a shift modulo 50 is a bijection, so it keeps equal numbers equal and different numbers different. The three copies of 43 are the fingerprint that gives n_0 away. After that, a sum taken modulo 50 pins down n_1.

Solution

Only n_0 and n_1 modulo 50 matter, so we take $0 \leq n_0, n_1 \leq 49$. Every a_k lies between 0 and 49, so each a_k is fixed once we know its residue modulo 50.

Removing the known shift

For the even positions, set $c_j = b_{2j} - j$ for $j = 1, \ldots, 6$. For the odd positions, set $d_j = b_{2j+1} - j$ for $j = 0, \ldots, 5$. Then

$$a_{2j} \equiv c_j - n_0, \qquad a_{2j+1} \equiv d_j - n_1 \pmod{50}.$$

From the coded message:

j	1	2	3	4	5	6
b_{2j}	27	36	35	40	37	48
c_j	26	34	32	36	32	42

j	0	1	2	3	4	5
b_{2j+1}	35	2	15	8	40	24
d_j	35	1	13	5	36	19

Finding n_0

Subtracting a fixed n modulo 50 is one-to-one. So two positions of the same parity hold equal a's exactly when their c's (or d's) are equal.

Ask yourself Where can the three 43s go?

The six d_j are all different, so the odd positions hold six different numbers, and at most one of them is a 43. So at least two 43s sit in even positions. Among the c_j the only repeat is 32, which occurs twice ($j = 3, 5$) and no more. So the even positions hold exactly two equal numbers, and these must be the two 43s, because 43 is the only number that appears more than once in the message. Hence

$$32 - n_0 \equiv 43 \pmod{50} \quad \Longrightarrow \quad n_0 \equiv -11 \equiv 39 \pmod{50}.$$

Now $a_{2j} \equiv c_j - 39 \equiv c_j + 11$, which gives

$$(a_2, a_4, a_6, a_8, a_{10}, a_{12}) = (37,\ 45,\ 43,\ 47,\ 43,\ 3),$$

since $42 + 11 = 53 \equiv 3$. All six are in the message, as they must be.

Finding n_1

Removing these six numbers from the message leaves the odd-position numbers

$$S = \{1,\ 7,\ 23,\ 24,\ 39,\ 43\},$$

so $\{d_j - n_1\}$ must equal S modulo 50.

Ask yourself How can we get n_1 without trying all fifty values?

Add up both sides. If the two sets agree, their sums agree modulo 50:

$$\sum d_j - 6n_1 \equiv \sum_{s \in S} s \pmod{50}, \qquad \text{that is,} \qquad 109 - 6n_1 \equiv 137.$$

So $6n_1 \equiv -28 \equiv 22 \pmod{50}$. Dividing through by 2, which also divides the modulus, gives $3n_1 \equiv 11 \pmod{25}$. Since $3 \times 17 = 51 \equiv 1$, we get $n_1 \equiv 11 \times 17 = 187 \equiv 12 \pmod{25}$. So $n_1 = 12$ or $n_1 = 37$.

The sum condition is necessary but not sufficient, so we test both values. With $n_1 = 37$, $d_0 - 37 = -2 \equiv 48$, which is not in the message, so $n_1 = 37$ fails. With $n_1 = 12$,

$$(a_1, a_3, a_5, a_7, a_9, a_{11}) \equiv (23,\ -11,\ 1,\ -7,\ 24,\ 7) \equiv (23,\ 39,\ 1,\ 43,\ 24,\ 7),$$

which is exactly S. So $n_1 = 12$.

The message

Interleaving the two halves, the original message is

$$23,\ 37,\ 39,\ 45,\ 1,\ 43,\ 43,\ 47,\ 24,\ 43,\ 7,\ 3,$$

with $n_0 \equiv 39$ and $n_1 \equiv 12 \pmod{50}$. The argument shows this answer is the only one: n_0 was forced, and only one of the two candidates for n_1 works.

Checks

Re-encode the answer. For example, $b_{12} \equiv a_{12} + 39 + 6 = 48$ and $b_7 \equiv a_7 + 12 + 3 = 58 \equiv 8$, both correct. A computer that tries all 2500 pairs (n_0, n_1) finds that exactly one pair, $(39, 12)$, decodes to the given collection of numbers, and that it gives the message above.

Common slips

- Getting the indexing wrong. The odd positions start at b_1 with $j = 0$, but the even positions start at b_2 with $j = 1$.
- Dividing $6n_1 \equiv 22 \pmod{50}$ by 6 as if 6 were invertible. It isn't, since $\gcd(6, 50) = 2$, and the congruence has two solutions modulo 50.
- Finding a value of n_1 that works without showing that no other value does. The question asks you to explain your method, so uniqueness needs an argument.

Beyond STEP

This is a Caesar (shift) cipher, and the attack is a small case of frequency analysis: a shift preserves the pattern of repeated letters. In a first course, the step from $6n \equiv 22$ to two solutions is the theory of linear congruences $ax \equiv b \pmod{m}$, which are solvable exactly when $\gcd(a, m)$ divides b. Affine ciphers $x \mapsto ax + b$ and, later, RSA are built on the same ideas. Davenport's *The Higher Arithmetic* covers congruences beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.